Educational Research Review 36 (2022) 100454

Contents lists available at ScienceDirect

Educational Research Review

journal homepage: www.elsevier.com/locate/edurev

Review

A meta-analytic review on incorporating virtual and augmented reality in museum learning

Yuting Zhou [a], Juanjuan Chen [a,*], Minhong Wang [b,c]

[a] College of Education, Zhejiang University, Hangzhou, China
[b] KM&EL Lab, Faculty of Education, The University of Hong Kong, Hong Kong, China
[c] Department of Educational Information Technology, East China Normal University, Shanghai, China

ARTICLE INFO

Keywords:
Augmented reality
Virtual reality
Informal learning
Systematic review
Meta-analysis

ABSTRACT

Virtual reality (VR) and augmented reality (AR) have been increasingly used to support museum learning by creating engaging and appealing learning experiences. However, there is a lack of meta-analytic reviews of empirical studies in this field. This study reviewed 51 relevant studies to investigate (a) the situations in which AR and VR have been applied in museum learning (RQ1) and (b) how AR and VR are incorporated in museum learning (RQ2), and conducted a meta-analysis of 17 studies to examine the effects of these technologies on learning achievement (RQ3). The results reveal that AR and VR have been mostly used in science, arts, and history museums to support the learning of science and art with a focus on conceptual knowledge. Second, they were often used to superimpose supplementary materials onto physical exhibits, dynamically visualize complex phenomena or concepts, and simulate virtual exhibition and narrative scenarios. Mobile devices were more commonly used than head-mounted displays (HMDs) and others. Third, AR and VR have significant positive effects on academic achievement (ES = 0.45) and perceptions (ES = 0.59) in museum learning. A number of suggestions for future research arose from this review.

1. Introduction

Since the 19th century, educators have begun to see museums as important informal learning environments that provide the general public (from children to adults) with visual, accessible and even tangible objects and exhibits to explore disciplinary content knowledge about art, history, culture, and science in depth, as well as to develop children's interest and career aspirations related to these subjects (De Rijcke & Beaulieu, 2011; Kisiel, 2006; Lewalter et al., 2021). Museums may house exhibits, collections or experiences that are often designed explicitly to meet educational goals targeting school-aged children (Crowley et al., 2014). However, museum educators often face the challenge of creating engaging activities for learners (Watson & Werb, 2013). In particular, guided school field trips to museums are mostly based on a whole-class lecture model that constrains the group's movements to exhibits and displays preselected objects. These trips generally focus more on meeting formal educational goals valued in schools, but fail to foster lifelong learning and social skills valued in museums, such as inquiry, observation, and conversational elaboration on topical issues in science, art, and history (Kisiel, 2003).

* Corresponding author. College of Education, Zhejiang University, Room 616, Block 10, Chengjun Building, 866 Yuhangtang Road, Hangzhou, 310058, China.
E-mail address: jjchen101@hotmail.com (J. Chen).

https://doi.org/10.1016/j.edurev.2022.100454
Received 8 November 2021; Received in revised form 22 April 2022; Accepted 23 April 2022
Available online 29 April 2022
1747-938X/© 2022 Elsevier Ltd. All rights reserved.

Virtual reality (VR) and augmented reality (AR) technologies are among the new technologies that are increasingly integrated into museum educational activities and aim to create more engaging and appealing learning experiences (Crowley et al., 2014). VR is a computer-generated completely virtual and simulation environment that provides a sensory illusion of being present in another environment (Makransky & Lilleholt, 2018; Renganayagalu et al., 2021; Smith, 2015). Users experience deep immersion, presence, and interactivity through specific VR technology devices like helmets or glasses on which the scene is represented. VR has been broadly utilized in education and has shown promising effects on learners' cognition (Akçayır & Akçayır, 2017; Morélot et al., 2021), behavioral skills (Çakiroğlu & Gökoğlu, 2019), and positive emotions such as enjoyment (Alfadil, 2020; Innocenti et al., 2019). In particular, VR technologies have been used to enrich learning experiences in museums. For example, VR can be used to reconstruct virtual historical settlements where visitors are totally surrounded by the scenes of ancient society and thus foster a holistic understanding of ancient society, which is difficult to conceive using only the artifacts and annotations shown in museums (Puig et al., 2020).

Unlike VR, which immerses users in a totally virtual and simulation environment, AR is a combination of virtual space and physical scenes (Carmigniani et al., 2011; Garzón et al., 2019), i.e., overlaying real-world objects or their digital representations with extra text, audio, video, or other virtual elements or information which are known as "digital augmentations" (Azuma, 1997; Sommerauer & Müller, 2014). This superimposing digital augmentation effect often occurs through a camera view with mobile platforms or a see-through display such as smart glasses (Pellas et al., 2019; Scavarelli et al., 2021). For example, a mobile device camera captures and recognizes a real-world object and then overlaps this object or its digital representation with extra text, audio, videos or other virtual elements (Calle-Bustos et al., 2017). In the study Yoon et al. (2017), when a physical ball is floating in the front of a Bernoulli blower, an AR device presents a digital image of the moving air around a digital ball that has been generated by the AR device as a digital representation of the physical ball; that is, visitors can see both the physical ball and the digital ball with a superimposed image of moving air. In other situations, AR devices directly project digital virtual objects in the physical space, e.g., displaying a virtual volcano in the classroom. Therefore, in museum learning, AR can provide learners easy access to unobservable phenomena or digital information on exhibits and promote their understanding of the real world (Danaei et al., 2020; Ibanez & Delgado-Kloos, 2018).

The theoretical foundation of incorporating VR and AR in museum learning is grounded in experiential learning (Behrendt & Franklin, 2014; Kolb, 2014), situated learning (Falk & Dierking, 2000; Johnson, 2002), active learning (Prince, 2004), and the multimedia learning theory (Mayer, 2014; Mayer & Estrella, 2014). Experiential learning typically involves four interactive learning processes: concrete experience, reflective observation, abstract conceptualization, and active experimentation (Kolb, 2014). Learners gain deeper understanding of the learning contents through these learning processes in authentic and first-hand context (Behrendt & Franklin, 2014; Kolb, 2014). Similar to experiential learning, situated learning involves learning within the context in which knowledge is situated and it might be used and applied (Choi & Ahn, 2021; Johnson, 2002). Active learning refers to learners' active engagement in meaningful learning activities and their reflection on learning (Chiang et al., 2021; Prince, 2004). Multimedia learning theory emphasizes the presentation of text and relevant graphical illustrations to promote learners' cognitive processing. Recently, researchers have expanded multimedia learning theory by incorporating emotional designs that can motivate learners to exert more mental effort to make sense of the presented material (Mayer, 2020; Mayer & Estrella, 2014). There is no doubt that VR and AR technologies are able to combine multiple media contents (such as texts, audio, video), provide an interactive, engaging educational context to support experiential and situated learning in museums, and induce learners to search more information for meaning-making and understanding, thus promoting active learning (e.g., inspecting historical settlements of ancient society) (Huang et al., 2016; Sultan et al., 2019).

1.1. Literature review

1.1.1. Museum learning

According to the International Council of Museums (ICOM), museums are institutions open to the public that acquire, conserve, research, communicate and exhibit the tangible and intangible heritage of humanity and its environment for the purposes of education, study and enjoyment (ICOM, 2007). They involve various objects, exhibits and programs related to the subjects of science, history, archaeology and the arts (Andre et al., 2017). Learning in museums (also called museum learning) enables visitors to understand the value of historical artifacts, learn about science, history and the arts and respect different cultures (Guzin et al., 2017; Xanthoudaki, 1998). Moreover, museum learning has shown great promise in increasing interest and cognitive skills (Yoon et al., 2012).

However, museums are currently suffering from reduced interest and lower attendance among the younger generations (Crowley et al., 2014). Thus, it is important to improve visitors' interest by designing engaging learning experiences (Hassan & Ramkissoon, 2016). Existing review studies on museum learning have focused on the adoption of technology in museums. Wang et al. (2017) reviewed the studies on ubiquitous language learning using location-based systems in museums from the perspective of system usefulness, activity usefulness and activity playfulness; they found that ubiquitous location-based systems using mobile devices helped to promote students' collaboration in group learning activities. Lin et al. (2021) investigated the trends in research on museum-based mobile learning in terms of research methods, learning domains, location-based technologies, learning strategies, and research topics. Xu et al. (2021) conducted a meta-analysis analyzing 42 experimental or quasi-experimental studies on technology applications in museum learning published during 2011–2021 and showed that technology application has a moderate or high effect on museum learning. In recent years, AR and VR are increasingly used to create engaging activities for learners (Falomo Bernarduzzi et al., 2021; Scavarelli et al., 2021).

1.1.2. AR and VR in museum learning

Many empirical studies have explored the use of AR and VR in museum learning across different domains, such as science, art,

archaeology, medicine and the military. For example, Oh et al. (2018) designed a multiuser participatory simulation called ARfract, where visitors can learn about complex concepts such as the refraction of light through full-body immersion using optical see-through AR glasses, projection-based AR, and gesture technology. Yoon et al. (2017) used an AR device to help students develop a more accurate understanding of fluid dynamics, a challenging concept of Bernoulli's principle in a science museum by visualizing air flow. Likewise, in art museums, history museums or archaeology museums, AR and VR have been widely exploited to provide augmentation or simulation of artworks and relics to create a more vivid and immersive learning environment. For example, in Chang et al. (2014), a mobile AR guide system presented paintings in a more instinctive way, aiming to foster college students' appreciation for painting. Moreover, AR has also been used in medical museums to display virtual labels, images and natural interfaces to advance students' understanding of medical specimens (Sugiura et al., 2019).

The learning outcomes measured in existing studies vary and generally include knowledge achievement, thinking skills, individuals' affections, etc. For example, Yoon et al. (2017) found that even when given only a short period for investigation in a science museum, students using AR demonstrated significantly greater knowledge gains than those who did not. Some studies have reported that AR/VR-assisted museum learning promotes learners' higher-order thinking skills, such as creative thinking (Guazzaroni, 2013; Hammady et al., 2020), inquiry (Hsiao et al., 2016), and critical thinking skills (Poce et al., 2019). Some studies have investigated learners' motivation and emotions. For example, Dieck et al. (2018) and Puig et al. (2020) revealed that these technologies enhance visitors' satisfaction and enjoyment and that wearable equipment helps to personalize their learning experience. Additionally, Nechita and Rezeanu (2019) reports that a multisensory augmented museum space can enhance empathy by allowing learners to experience various historical circumstances from a first-hand perspective.

However, other studies found negative effects of AR/VR-assisted museum learning. For example, Savela et al. (2020) found that AR games in a science center did not necessarily improve learning performance or social interaction when compared to the traditional pen-and-paper learning method. Additionally, learners claimed that using VR devices made them feel nauseous and dizzy (Rhee, 2019), and some devices such as HMDs also physically inconvenienced visitors (Oh et al., 2018; Sugiura et al., 2019). Therefore, whether and to what extent AR and VR applications could benefit visitors in museum learning are questions that still need to be synthesized and clarified.

1.1.3. Prior reviews on AR and VR in education

AR and VR have become a focus in educational studies in recent years, most of which concentrate on formal learning environments. Akçayır and Akçayır (2017) presented a systematic review on AR application in both K-12 and higher education and explored the advantages and challenges of using AR, concluding that AR can potentially support learning and teaching. Bacca et al. (2014) reviewed the usage, advantages, limitations, effectiveness, challenges, and features of AR in educational settings and found that it has been most commonly used in science, humanities and arts learning. Radianti et al. (2020) reviewed the application of VR in higher education by focusing on three key aspects: the learning contents, the design elements, and learning theories. Saltan and Arslan (2017) provided a comprehensive overview of the emergence of AR research on pedagogical and educational outcomes, specifically in formal education, and illustrated the evidence of improved academic achievements and perceptions. Goff et al. (2018) reviewed AR in exhibit-based informal science education settings and reported the STEM-focused topics covered by AR applications, various devices, and positive learning outcomes.

The abovementioned reviews provide syntheses of the trends, advantages and challenges of using AR and/or VR in educational settings, mostly in formal settings. Thus, there is as yet no meta-analytic review of empirical studies on the use of AR and VR in museum settings. Specifically, it is unclear in which contexts these technologies have been applied in museum learning and how the learning activities incorporating AR and VR are designed (i.e., the design elements). Moreover, the effects of AR and VR on museum learning are ambiguous as some reported positive effects while others reported negative effects.

1.2. The current study

Thus, the present study aims to provide a comprehensive meta-analytic review of AR/VR-assisted museum learning. First, education researchers and museum practitioners may be interested in the contexts in which AR and/or VR have been used, such as the learning domains and learning contents. Second, it is important for them to gain a deeper understanding of AR and VR affordances and their corresponding design elements. Third, there is a great need to know the overall effects of AR and VR on museum learning. Therefore, our research questions are as follows:

Research Question 1 (RQ1): In which contexts or situations have AR and VR technologies been applied to support museum learning?

Research Question 2 (RQ2): What are the design elements (the technology affordances, in particular) of AR and VR for museum learning?

Research Question 3 (RQ3): What are the effects of AR/VR-assisted museum learning versus non-AR/VR-assisted museum learning on students' academic achievement and perceptions?

To answer RQ1, we analyzed museum types, learner types, learning domains, and learning contents. To address RQ2, we explored the design elements of existing empirical studies, the technology affordances, and how the design elements used therein were implemented in technology devices. Finally, we conducted a meta-analysis to investigate the impact of AR and VR on museum learning (RQ3).

2. Method

2.1. Literature search

The studies were located through a comprehensive literature search of the Web of Science (WOS), Scopus, IEEE Xplore, and ProQuest online databases using the search query: ("augmented reality" OR "virtual reality" OR "mixed reality" OR "MR" OR ″3D″ OR "VR" OR "AR") AND ("science center" OR "museum") AND ("learn*" OR "educat*" OR "train*" OR "teach*"). The reason for including MR is that MR encompasses a mixture of reality with elements and features of both AR and VR. The last search was conducted on March 17, 2021, with the end date being December 31, 2020 and no limit for the start date. For WOS and Scopus database, document type was article and book chapter; for IEEE Xplore, no restriction was applied, and for ProQuest database, document type was journal articles, dissertations and conference proceedings. Both the English and Chinese language articles were included. The initial database search yielded 893 records, of which 268 were from the Web of Science, 338 were from Scopus, 163 were from ProQuest, and 124 were from IEEE Xplore. These records were imported into the literature management tool EndNote and were checked for duplicates. After duplicates were removed, 718 records remained for further filtering based on the following inclusion and exclusion criteria.

2.2. Criteria for inclusion and exclusion

To be included, studies had to meet all of the following criteria:

1. The research must be an empirical study, which means the study collects and analyzes empirical data; thus, secondary data analyses, meta-analyses, theoretical papers, and simulated results (from simulation models) were excluded.
2. The research environment must be a museum or science center; thus, studies carried out in cultural heritage, simulated outdoor learning environments and school classrooms were excluded.
3. AR and VR technologies must be clearly identified in the study; thus, studies that only used 3D models, 3D printing techniques, or online virtual museums via computer desktops were excluded.
4. The study must include assessments of learning effectiveness; therefore, studies reporting only the evaluation of AR and/or VR from the technology implementation perspective are excluded.
5. The full text of the literature must be available.

Moreover, there were additional criteria for RQ3:

6. The research must use a controlled experimental or quasi-experimental design, i.e., employing AR/VR-assisted museum learning in experimental groups and non-AR/VR-assisted museum learning in control groups.
7. The results must provide sufficient data for calculation of effect size (ES) of an intervention, including means and standard deviations and errors (SD and SE), F ratios, t values, chi-square statistics, r-indices, or z score of Mann-Whitney's *U* test, as well as the number of participants in the experimental and control groups. Thus, studies with insufficient information for the calculation of ES were excluded for RQ3.

2.3. Literature screening and selection process

The processes of literature filtering followed the recommendations of the PRISMA framework (Moher et al., 2009), as depicted by

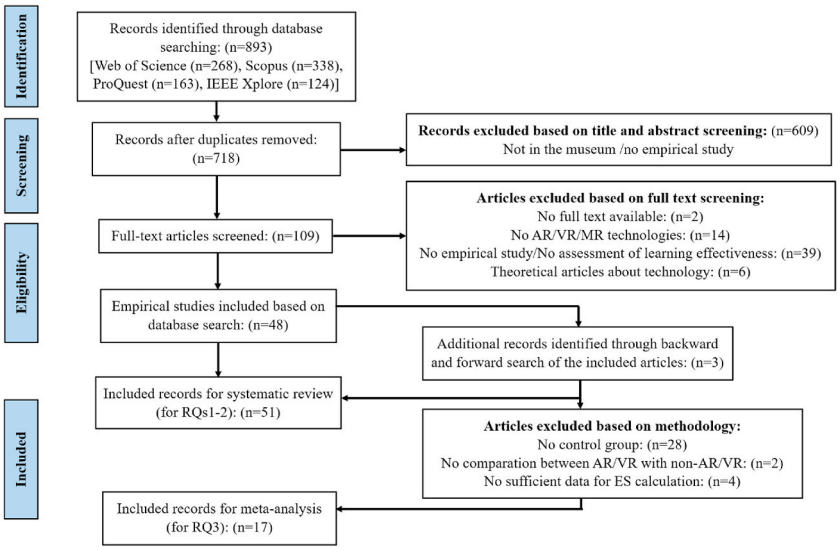


Fig. 1. PRISMA flow diagram.

the flow diagram (Fig. 1). First, we excluded 609 records by screening the titles or abstracts because they either did not involve learning in museums or were not empirical studies. The remaining 109 records were checked for full-text availability, and 2 records were excluded due to the unavailability of the full text. Another 59 records were excluded based on a full-text screening. Forty-eight records met inclusion criteria 1–5 for RQ1 and RQ2. Then, backward and forward searches of the 48 records' reference list were conducted. With this additional search, we identified another 3 empirical studies that met inclusion criteria 1–5. For RQ3, 34 records were excluded for not meeting criteria 6 and 7. Thus, 51 records were included in the systematic review to answer RQ1 and RQ2, 17 of which were for a meta-analysis to address RQ3.

Two of the researchers first reached an agreement on the inclusion and exclusion criteria. Then, they independently reviewed the initial search results, with an agreement coefficient of 0.88. They discussed the conflicts and reached a consensus. For the selected articles, they discussed the coding framework, which was confirmed by the other researchers.

2.4. Coding framework

To describe and differentiate selected studies and extract findings, the coding framework was established from diverse dimensions, including publication time and area, learner types, museum types, domains or subjects, learning contents, design elements or technology affordances, technology devices, and learning outcomes.

2.4.1. Codes for museum types, learner types and learning domains

The codes for the museum types are based on the museums where the experiment is conducted, including science museum (or science center), art museum, history museum, natural history museum, archaeological museum, medical museum, military museum, and not specified. Learner types are divided into five subcategories: K-12 learners (primary and secondary school students), higher education learners (college, university, or postgraduate students), adults (i.e., non-student learners), family (parent(s) with children) and the general public (ordinary museum visitors without a particular type). The codes for learning domains include biology, mathematics, physics, medicine, botany, art, history, archaeology, meteorology, and not specified.

2.4.2. Codes for learning content

The codes for the learning content are given in Table 1 and reference the coding framework of Radianti et al. (2020). Five categories of learning content emerged from the 51 selected articles.

2.4.3. Codes for design elements or technology affordances of AR and VR in museum learning

With respect to how AR and VR are employed to support museum learning, their technology affordances are classified into 7 categories, as listed in Table 2.

2.4.4. Codes for devices of AR and VR technologies

Based on the classification of AR and VR devices proposed by the existing literature (e.g., Arici et al., 2019; Radianti et al., 2020), this study coded the technology devices in which AR/VR have been implemented into five categories, as shown in Table 3.

2.4.5. Codes for learning outcomes

The codes for learning outcomes were based on the work by Chen et al. (2018) and include academic achievement and perceptions. Academic achievement includes the knowledge and skills acquired through museum learning, e.g., conceptual knowledge and cognitive skills. Perceptions refer to self-reported feelings of the learning experience (Kyndt et al., 2013), which are often measured by surveys or questionnaires, including perceived learning, enjoyment, engagement, satisfaction, motivation, novelty, exploration and perceived challenge.

Table 1
Coding framework for the learning content.

Categories	Explanation
Declarative knowledge	The use of AR and VR intends to help learners to remember factual knowledge and develop conceptual understanding (e.g., theoretical concepts and physical principles).
Analytical and problem-solving skills	The use of AR and VR intends to improve learners' analytical skills (e.g., analyze the factors affecting weather changes).
Emotional experience	The use of AR and VR can provide learners with empathic experience (e.g., experiencing AR-constructed historical battles of ancient Egypt).
Behavioral impacts	The use of AR and VR aims to change the behavior of visitors (e.g., presenting harmful impacts of smoking).
Procedural-practical knowledge	The use of AR and VR provides learners with interactions or animations to acquire procedural or practical knowledge (e.g., sculpturing).
Others	Articles that could not be classified into the alternative concepts above
Not specified	There is no statement or implicit information about the learning content of AR and VR usage.

Table 2
Coding framework for the design elements and affordances of AR and VR for museum learning.

Categories	Explanation	Examples	Technologies
Manipulation of virtual objects	Users select or interact with virtual objects in order to change their position or state.	Throwing virtual objects to the right place in a VR game.	AR, VR
User-generated content	Users generate new content with AR and VR applications and the newly generated content can be uploaded, saved and shared with other users.	Developing an interactive resource or a digital image narrative about the learning content.	AR, VR
Quiz game-based learning task	Users complete AR and VR game quizzes or questions related to exhibitions.	Discovering the virtual floating questions around virtual exhibits and answering them.	AR
Simulation of exhibition	The objects or collections of the exhibition are completely generated by AR and VR applications.	Virtual diorama of real flowers displayed on the screen covering the real floor.	AR, VR
Simulation of invisible phenomenon	AR and VR simulate or dynamically visualize invisible phenomena after visitors manipulating museum devices, mainly visualizing abstract concepts in physics.	Digitally displaying virtual arrows of moving air to explain the relation between the moving air speed and pressure.	AR
Simulation of narrative scenario	AR and VR reconstruct the virtual historical scenes with high graphic resolution, replicating some narrative scenarios of the past.	Neolithic settlement in a computer-generated virtual environment.	AR, VR
Superimposing supplementary materials	AR and VR provide superimposed information in various forms on augmentation layer to explain the exhibition content.	Superimposing texts, pictures, or videos on exhibits captured by camera, or in the form of a virtual human guide.	AR

Table 3
AR and VR devices.

Categories	Explanation	Technologies
Mobile device	AR technology is implemented with portable, wireless computing devices that can be handheld, including mobile phones and tablets.	AR
Head-mounted display (HMD)	AR and VR technologies are implemented with a device worn on the head or built in as part of a helmet containing small computer and monitor, including HTC VIVE, Google Cardboard, Microsoft HoloLens, etc.	AR, VR
Desktop device	AR and VR technologies are implemented with desktop computers and digital devices that are not movable in museums.	AR, VR
Projector	AR and VR technologies are implemented with a device for projecting photographs or videos onto a screen or floor in museums.	AR, VR
Not specified	There is no statement or description of AR and VR devices.	AR, VR

2.5. Inter-coder reliability

The first two authors discussed the coding framework, which was approved by other authors of this study. Two trained coders independently performed all of the coding. Cohen's *k* was run to determine the degree of inter-coder reliability on each code. Cohen's *k* was 0.96 for museum type, 0.98 for learner type, 0.96 for learning domain, 0.98 for learning content, 0.90 for design element, and 0.96 for technology device. All disagreements were resolved through further discussion.

2.6. Meta-analysis

The meta-analysis methods used in this study follow the meta-analysis procedures proposed by Lipsey and Wilson (2001). The random-effects model was used, considering the variability of contexts, participant age, subject domains, and so on. First, effect size (ES) was extracted from studies. For studies with means and SDs, Cohen's d was calculated for the indices of ES. Cohen's d usually uses the standardized mean difference between the experimental and control conditions divided by the pooled standard deviation (SD) and is commonly represented by Formula 1.

$$d = (M_1 - M_2) \Big/ \sqrt{S_1^2(n_1 - 1) + S_2^2(n_2 - 1)/n_1 + n_2 - 2} \tag{1}$$

where M_1 and M_2 represent the mean score of the experimental and control conditions, respectively, when only posttest data are reported or the gain score when both pre- and posttest scores are available (meaning posttest minus pretest scores) for the variables of interest (e.g., academic achievement, perceptions). S_1^2 and S_2^2 represent the posttest standard deviation of the experimental and control conditions, respectively, and n_1 and n_2 are the number of participants in both conditions, respectively.

Borenstein et al. (2009) suggested that Hedges' g can reduce the small-sample size bias. Thus, Cohen's d was converted to Hedges' g using Formula 2.

$$g = d\left(1 - \frac{3}{4(n_1 + n_2) - 9}\right) \tag{2}$$

When the raw data in one study were in the forms of a *t*-test, F test, r index, or chi-square values, the effect size d was calculated via

algebraically equivalent formulas provided by Lipsey and Wilson (2001) or a web-based ES converter developed by them.

When multiple measurements of one outcome were reported (e.g., knowledge gains and skill acquisition for academic achievement), their effect sizes were averaged to guarantee only one ES for each outcome, thus ensuring the independence of extracted ESs (Lipsey & Wilson, 2001).

When there was more than one experimental condition and only one single control condition or there was more than one control condition and only one experimental condition, dependent ESs for each comparison were averaged with a weight by sample size (Tanner-Smith & Tipton, 2014).

3. Results

In total, 51 studies were included to address RQs 1 and 2, 17 of which were identified to conduct a meta-analysis answering RQ3. Note that the number in parentheses indicates the number or percentage of selected studies.

3.1. General characteristics of the selected articles

The articles selected were published between 2012 and 2020: 2 in 2012 (3.92%), 3 in 2013 (5.88%), 5 in 2014 (9.80%), 3 in 2016 (5.88%), 5 in 2017 (9.80%), 10 in 2018 (19.61%), 12 in 2019 (23.53%) and 11 in 2020 (21.57%). There were no more than five articles for each year between 2012 and 2017 and approximately 10 articles per year between 2018 and 2020. The results indicate that there are increasing research interests in the application of AR and VR technologies in museum learning.

The first authors of the included articles were mostly from the United States (11 articles), Taiwan (7 articles) and the United Kingdom (7 articles). The other 15 articles were published by authors from other European countries.

Of the 51 selected publications, there are 32 journal articles (62.75%), 11 conference papers (21.57%), 6 book chapters (11.76%), and 2 dissertations (3.92%). As for the publication languages, there was only one article in Chinese, with the other 50 in English.

3.2. Contexts or situations using AR and VR in museum learning

To address RQ1, this review synthesized the results in three dimensions: learner types, museum types, and learning domains.

3.2.1. Learner types

Regarding the participants in the included articles (see Fig. 2), most were K-12 students (40%), followed by the general public (34%) who are daily museum visitors. Other studies focused on higher education students (17%) and adults (7%). Note that there were 2 studies involving both university and K-12 students (Bossavit et al., 2018; Nechita & Rezeanu, 2019). Some researchers were also concerned with the conversations among family members (Wang, 2014).

3.2.2. Museum types

As illustrated in Fig. 3, most of the studies (18, 35.3%) were conducted in science museums (including science centers), and the technologies involved were all AR technologies. Some studies focused on using AR and VR in art museums (13, 25.5%), history museums (14, 27.4%), and archaeology museums (4, 7.8%). There was 1 study using AR in a medical museum, which incorporated AR to deepen learners' understanding of medical specimens (Sugiura et al., 2019). Note that there were two studies that incorporated both AR and VR technologies within one history museum (Bell & Smith, 2020; Jung et al., 2016). Moreover, AR has been more commonly used in multiple types of museums, whereas VR in the included studies has only been incorporated into art, history, and archaeology museums. In summary, AR and VR technologies are most often used in science, history and art museums to support learning.

3.2.3. Learning domains

As shown in Fig. 4, AR and VR were most often applied in art learning in museums (13 of 51 articles), among which 8 were AR

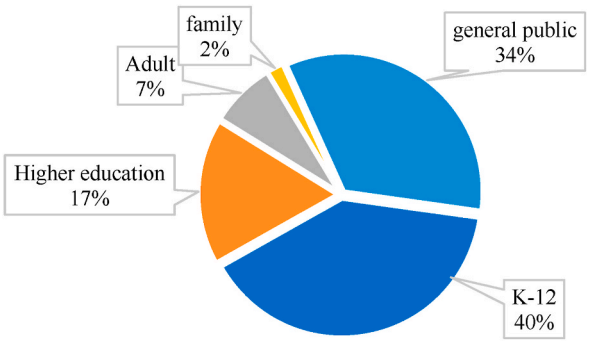


Fig. 2. Learner types.

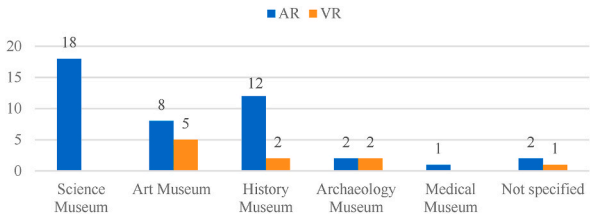


Fig. 3. AR and VR in different museums.

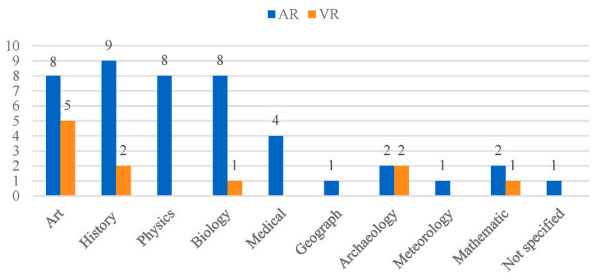


Fig. 4. Learning domains of AR and VR applications.

applications and 5 were VR applications. For art learning, AR and VR superimposed additional media content such as images, video and 3D information on art collections, thus presenting the talents of the artists in a new way and enriching the visitors' learning experience. AR and VR have also often been applied in the domains of history (11), biology (9), and physics (8). For history learning, these technologies vividly bring historical scenes to life, such as ancient Egyptian battles (Hammady et al., 2020) and life of ancient people settling in Romanian Brasov city (Nechita & Rezeanu, 2019). For science learning, these technologies were mainly used to make the invisible observable by visualizing invisible phenomena or abstract concepts, such as the Doppler phenomenon and Boltzmann's molecule movement (Salmi et al., 2017). Some articles did not mention a specific domain (e.g., Haryani & Triyono, 2020; Jung et al., 2016; Moorhouse et al., 2019). More details about the learning topics in selected empirical studies can be found in the supplementary file. Note that in Jung et al. (2016), two history learning activities were designed with AR and VR technologies; in Bell and Smith (2020), participants learned biology and mathematics with the help of both AR and VR.

3.2.4. Learning content

Fig. 5 shows that AR and VR were most frequently used to promote declarative knowledge learning in museums (33), such as basic knowledge of rheumatoid arthritis (Kosa et al., 2019), Bernoulli's principle (Wang, 2014; Yoon et al., 2017), or ancient civilizations (Khan et al., 2020). They were also used to foster thinking skills, such as analytical and problem-solving skills (5) in game or adventure situations (Guazzaroni, 2013; Hsu et al., 2018; Poce et al., 2019), or analyzing the internal structure of a baleen whale (Takahashi et al., 2013, pp. 385–390). There were only 2 studies on learning procedural-practical knowledge such as sculpturing (Koutsabasis & Vosinakis, 2018) or using fishermen's tools (Sulaiman et al., 2019, pp. 79–83), 2 studies on promoting emotional experience by

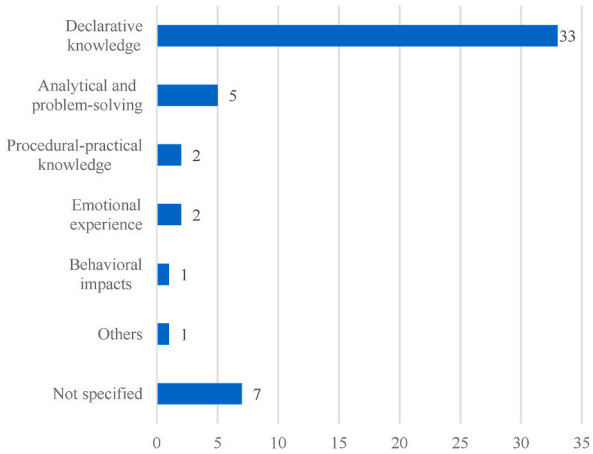


Fig. 5. Learning content.

engaging visitors in ancient Egyptian battle scenarios (Hammady et al., 2020) and surrounding visitors with highly emotional content related to the Nazi Holocaust (Marín-Morales et al., 2019), and 1 study on changing learners' behaviors by presenting children with realistic AR models of a smoker's lungs to ward children off this dangerous habit (Borovanska et al., 2020). The numbers for the categories *Others* and *Not specified* are 1 and 7, respectively.

3.3. Design elements or technology affordances of AR and VR for museum learning

To address RQ2, this review synthesized how AR and VR were employed to support museum learning (i.e., the design elements) and how their affordances were implemented with specific technology devices.

3.3.1. Design elements and technology affordances

The results for the categorization of design elements can be seen in the upper part of Fig. 6. Based on the coding framework presented in Table 2, 23 studies used AR or VR to superimpose supplementary information onto physical exhibits to explain the exhibition content (e.g., Tabone, 2020). The superimposed information was in the form of labels or texts (Juan et al., 2017, pp. 466–468; Tabone, 2020), images, audio and video (Damala et al., 2016), or 3D models (Borovanska et al., 2020; Connaghan et al., 2019; Ghouaiel et al., 2017). This could be seen as a similar function of labels used for traditional exhibits. Eight studies used these technologies to provide simulations or dynamic visualization of typically invisible phenomena, e.g., the travel and refraction of light (Oh et al., 2018) and the flow of electrical currents on the visitor's body (Yoon et al., 2012). When there were no physical objects or exhibits, AR and/or VR were used to simulate the exhibitions by showing real cascading flowers over the gallery floor (Harrington et al., 2019), or to simulate narrative scenarios such as the underwater archaeological world (Li & Chang, 2017).

In addition to augmenting visitors' view, one prominent affordance of AR and VR is to allow for user interaction with virtual objects. Among the selected studies, visitors were able to manipulate virtual objects (3), including manipulating 3D objects to observe weather variations with an AR app (Hsiao et al., 2016), performing quiz-based tasks (4), and generating content (4), such as constructing Cycladic figurine with stony tools used in ancient times (Koutsabasis & Vosinakis, 2018). In some studies (e.g., Dudzik, 2018; Jung et al., 2016; Puig et al., 2020), the design elements were more than one when an article contained two or more studies that used different design elements to support various learning activities.

3.3.2. Types of technology devices

The lower part of Fig. 6 shows the frequency of the devices appeared in the 51 articles. The most commonly used type of device was mobile devices such as smartphones and tablets (28). The second was HMD devices (15), including Google Cardboard, HTC VIVE, Microsoft HoloLens, etc. There were 6 studies using desktop devices and 5 studies using projectors. However, there were 2 studies that did not clearly describe which type of devices they used. Note that the total number of devices is larger than 51 because some articles used more than one device either to create different activities (Dudzik, 2018; Jung et al., 2016; Sugiura et al., 2019) or to provide a better learning context (Nechita & Rezeanu, 2019).

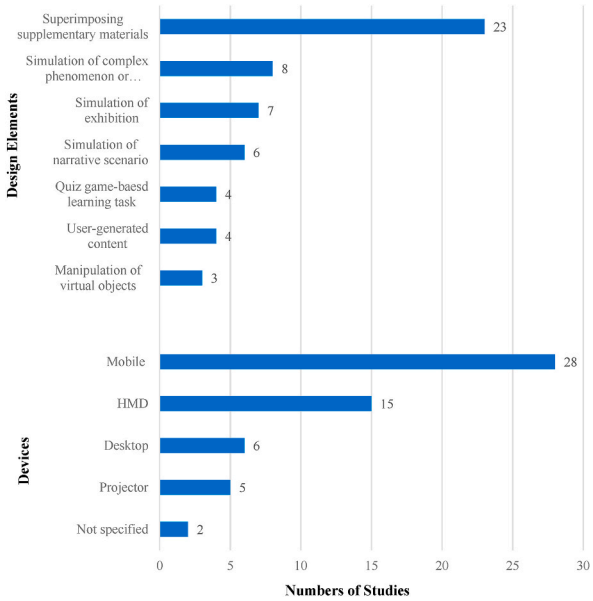


Fig. 6. Results on design elements and technology devices.

3.3.3. Links between design elements, AR and VR technologies, and specific devices

To provide a clear overview of how AR and VR affordances are implemented with different devices, bubble charts (Radianti et al., 2020) are presented in Fig. 7. The size of the bubbles represents the number of articles found for a given implementation. The device categories in the middle part of Fig. 7 can be interpreted in combination with both the left and right bubble charts. It should be noted that the number of articles in the bubble charts exceeds 51 because some articles may include two or more studies integrating different devices with design elements. For example, HMD was used to simulate a narrative scenario as well as an interactive game (Puig et al., 2020).

Links between design elements and specific devices. The right bubble chart of Fig. 7 presents how different design elements were implemented in specific devices. First, mobile devices were often used to superimpose supplementary information onto physical exhibits (19). Visitors could use smartphones or tablets to scan exhibits or QR codes or download applications to obtain exhibit-related knowledge in the form of digital text and multimedia information overlaid on physical objects. For example, visitors could view and listen to the corresponding commentary about paintings after capturing the paintings with smartphone cameras (Chang et al., 2014).

Second, simulations of invisible phenomena were implemented using several types of devices including desktops (4), projectors (2), and HMDs (1). Simulation of exhibitions without the presence of physical objects was implemented with HMDs (3) and mobiles (3). It is worth noting that the simulation of narrative scenarios was exclusively implemented with HMDs, e.g., HTC VIVE displaying the underwater archaeological virtual world and enabling visitors to gain a better understanding of sea creatures (Li & Chang, 2017).

Third, quiz game-based learning tasks were all performed on mobile devices (4). For example, in Savela et al. (2020), visitors used a cellphone AR app to visualize the virtual floating question marks around related exhibits and must tour the entire exhibition room to answer all the quiz questions.

Fourth, visitors generated content with mobile devices (1), projectors (1) or desktops (1). For example, visitors played the role of an ancient craftsman to create a figurine with a kinesthetic sculpturing application using a desktop device (Koutsabasis & Vosinakis, 2018).

Links between specific devices and AR and VR technologies. The left bubble chart in Fig. 7 depicts how a specific device was used (i.e., for augmentation or total immersion). It shows that mobile devices such as smartphones and tablets were mostly used for AR (28). Visitors can view digital augmentations through applications installed on phones or tablets (Borovanska et al., 2020; Connaghan et al., 2019).

HMDs were equally used for AR (8) and VR (8). For AR, HMDs were mainly low immersion headsets such as Google Cardboard and Google Glass (Tabone, 2020; Yu et al., 2019); that is, users could see the physical world rather than being totally surrounded with computers-generated images and sound. HMDs for VR, such as HTC VIVE or Samsung Gear VR Glasses (Baradaranrahimi et al., 2020, p. 1; Li & Chang, 2017; Puig et al., 2020), allowed users to experience a high degree of immersion and provided a complete sense of presence. Desktop devices (i.e., nonmovable computers or equipment in museums) were mainly used for AR (5) that displays digital augmentations after visitors interacted with exhibits (Yoon et al., 2018; Yoon & Wang, 2014).

The desktop device used for VR (1) is implemented with a handheld device called the Leap Motion Orion (Koutsabasis & Vosinakis, 2018). The projectors used for AR implementation (4) were either to give feedback (e.g., Dudzik, 2018) or to project physical phenomena in combination with desktops (e.g., Yoon et al., 2013). Projectors used in VR applications are implemented with a Kinect device to fully immerse learners (Bossavit et al., 2018).

3.4. The effects of AR/VR-enhanced museum learning on academic achievement and perceptions

A total of 17 studies met all of the inclusion criteria for RQ3 and examined the effects of AR/VR-assisted versus non-AR/VR-assisted learning on visitors' individual academic achievement and perceptions. Table 4 presents detailed information to address RQ3 and the 95% confidence interval for each outcome. Note that only 1 out of the 17 studies investigated the effects of VR with non-VR supported museum learning; thus, the effect size of AR and VR were not calculated separately.

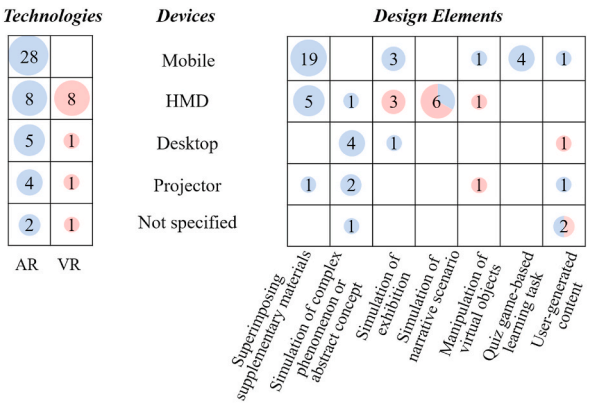


Fig. 7. Relationships between technologies, specific devices, and design elements.

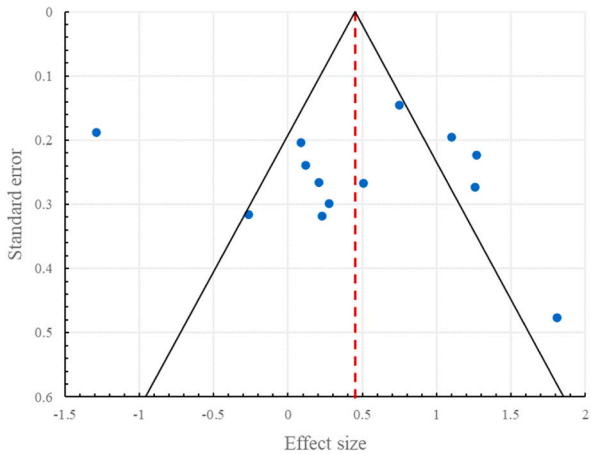
Table 4
Overall weighted mean effect sizes for effects of AR and VR in museum learning.

Participants	N	Outcome	k	$\bar{g}$	95% CI
1601 (N_{exp} = 827, N_{ctrl} = 774)	17	Academic achievement	13	0.45	[0.00, 0.89]
		Perceptions	12	0.59	[0.29, 0.91]

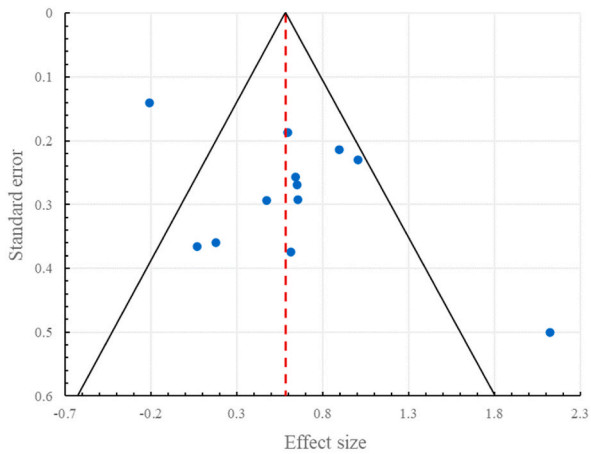
Note. N = total number of included studies; k = number of studies analyzed for each learning outcome; $\bar{g}$ = weighted mean effect size; CI = confidence interval.

3.4.1. Academic achievement

The weighted mean effect size for academic achievement synthesized from 13 studies was 0.45 (95% CI [0.00, 0.89], k = 13), indicating a significant positive effect of AR and VR on visitors' measured gains in knowledge and skills. In other words, visitors using AR and VR applications in museum learning had significantly better academic achievement than those who used non-AR and non-VR applications such as text or traditional audio guides. Individual effect sizes of the 13 primary studies ranged from −1.30 to 2.05. Sommerauer and Müller (2014), for example, examined the effect of AR in learning mathematical content through AR exhibitions. The experiment was conducted with 101 participants (from the general public) for 90 min in an anonymous national museum. All of the participants received hands-on training on how to use the mobile AR app and were randomly assigned to experimental or control groups. The results showed that participants in the experimental group outperformed their counterparts on questions related to AR exhibits, with an ES of 0.75.



A. Funnel plots with pseudo 95% confidence limits for academic achievement



B. *Funnel plots with pseudo 95% confidence limits for perceptions*

Fig. 8. Funnel plot of standard error by Hedges's g.

3.4.2. Perceptions

The weighted mean effect size for perceptions synthesized from 12 studies was 0.59 (95% CI [0.29, 0.91], $k = 12$), indicating a medium effect of AR and VR on visitors' perceptions. The individual effect sizes of the 12 primary studies ranged from −0.21 to 2.19. For example, Chang et al. (2014) implemented AR with a mobile device as a guide system for art appreciation in the Taipei Fine Arts Museum. The experiment was conducted among a sample of 131 college students. The participants in the experimental group who used AR-guided systems reported a better state of flow experience and were more proactive in appreciating the artwork than groups using audio only or groups without any guide.

The possibility of publication bias was evaluated in this meta-analysis by plotting a funnel plot (as shown in Fig. 8). For the measure of both academic achievement (Fig. 8A) and perceptions (Fig. 8B), the funnel plot suggests there was no publication bias.

4. Discussion

4.1. General characteristics of the selected articles

The meta-analytic review shows that the application of AR and VR in museum learning has received increasing attention especially in the last three years. The commercial availability of AR and VR devices such as Oculus Rift and Microsoft HoloLens (Scavarelli et al., 2021) makes it easier for researchers to conduct AR/VR-based educational studies in different learning fields, including museums.

4.2. Contexts or situations using AR and VR in museum learning

Most studies were conducted on K-12 students. The statistical results are not surprising because museums always play important roles in these students' informal learning, support diverse activities and enrich their after-school life. However, there were no studies conducted on teachers, who might put forward suggestions for how to better integrate AR and VR with museum learning after experiencing the activities first-hand.

Regarding the learning domains, AR and VR technologies are commonly used in the learning of art, science, and history. Artworks, historical exhibits, collections and relics require a more immersive presentation to better engage people. In this regard, AR and VR technologies provide digital augmented visual complementarity to the details of artworks (Chang et al., 2014). Meanwhile, previous studies have suggested that learners can develop a better understanding of abstract and challenging scientific concepts (e.g., Bernoulli's principle) with the help of AR (Cai et al., 2013; Chiu et al., 2015). AR can visualize the invisible mechanisms that underlie complex phenomena (Yoon et al., 2017).

Regarding learning contents, most studies focused on of the development of declarative or conceptual knowledge. To some extent, the affordance of AR and VR to support learning is underestimated. AR and VR can provide simulation environments for visitors to acquire practical skills such as sculpturing, develop higher-order thinking skills, or enhance emotional engagement or experience. However, there are relatively few studies investigating how AR/VR-assisted museum learning can promote learners' acquisition of these outcomes.

4.3. Technology affordances of AR and VR for museum learning

The findings imply that AR technologies were used more often than VR technologies in museum learning. This may be largely due to the easier access to the necessary devices. Recent developments in mobile technologies such as smartphones and tablets with built-in cameras, global positioning system and internet access have made AR applications available to the general public (Sommerauer & Müller, 2014). In addition, AR devices are cost-effective and easy to use compared to VR devices (e.g., HMDs), which are not as prevalent in museums currently.

The technology affordances of AR and VR for museum learning include superimposing information onto physical exhibits or objects (23 studies), simulating complex phenomena or abstract concepts (8 studies), simulating exhibitions (7 studies), and simulating narrative scenarios (6 studies). Acquiring detailed information about exhibits such as historical artifacts and developing a better understanding of the material is the main purpose of visiting museums. However, it is not always possible to attach paper-based labels to the exhibits, e.g., observing the structures and development of tumors in a medical museum. In this regard, superimposing information onto physical exhibits makes it possible. In addition, providing digital simulations of phenomena, exhibitions, and narrative scenarios is ideal for immersive and experiential learning (especially in science museums) to foster scientific literacy (Crowley et al., 2014).

However, only a few studies have explored applying AR and VR to enhance visitors' interactions with exhibits or allow visitors to generate virtual content in immersive environments (e.g., Hsiao et al., 2016; Koutsabasis & Vosinakis, 2018). This implies that the application of AR and VR is mainly in low-immersion modes.

4.4. Effects of AR and VR on museum learning

The positive effects of AR/VR-assisted museum learning (ES = 0.45 for academic achievement; ES = 0.59 for positive perceptions) can be explained by the fact that AR and VR provide enriching and immersive learning settings, as experiential or situated learning have proved to promote learning (Kolb, 2014). Learners can be actively involved in cognitive processing when a coherent mental representation of their own experiences is constructed (Mayer, 2014). In addition, although learners in AR and VR experimental group

and non-AR/VR control group received nearly the same learning content, the dynamic visualization of AR and VR actually contains more information in many situations, which may result in learners' higher achievement. The findings on academic achievement are consistent with a previous review that suggested the effectiveness of AR technologies in general educational settings (Bacca et al., 2014).

The learning outcomes of museum visits encompass not only knowledge acquisition in a narrow sense but also visitors/participants changes in interests and beliefs (Schwan et al., 2014). It is also hoped that museum visits could change visitors' attitudes and even their behavior. The technical novelty of AR and VR may attract visitors, hold their attention and encourage them to persevere in learning, thus resulting in more positive perceptions (Bettelli et al., 2020).

4.5. Suggestions for future research

Based on the above findings, suggestions for the design of future research on AR/VR-assisted museum learning are as follows:

(1) Extend the learning content in museums. A majority of the current studies focused on declarative knowledge learning, such as general information regarding the artists and their works, while few were related to procedural-practical knowledge and distal outcomes such as analytical and other thinking skills. Therefore, this learning content could be the learning objectives of future studies.
(2) Provide real immersive contexts. The design of museum learning environments should make full use of VR technologies to provide real immersive learning environments. Instead of merely providing animation on desktop devices with low immersion, the learning settings should make more use of VR technologies. This requires collaboration between museum education designers and technological experts.
(3) Take into account the cognitive levels of visitors at different ages. The cognitive abilities and usage of technologies and devices vary among visitors of different ages, and their preferences concerning museums visits are also diverse. Thus, future research should consider the differences among visitors to meet different visiting requirements.
(4) Employ controlled experimental design. It is typical that many of the selected 51 empirical studies focused on the usability of AR and VR technologies and were conducted among a single group of participants without controlled experiments. In this review, only 17 out of the 51 selected articles used a controlled experimental design. More experimental or quasi-experimental studies with a control group design are needed to investigate the effects (such as the acquisition of knowledge and skills) of using AR and VR technologies in museum learning.

5. Limitations

This study is subject to several limitations. First, this review examined articles from four databases: Web of Science, Scopus, IEEE Xplore and ProQuest. However, some relevant studies are not included in these four databases. Second, this study has focused on traditional museums. There are various informal learning environments, such as zoos, aquariums, arboretums and historical sites, which have expanded the concept of the "museum". AR and VR applied in such types of museum learning were not included in this study. Thus, future research should include these types of museums to provide a more complete view of the impact of AR and VR on museum learning. Third, this review does not examine the barriers to the usage of AR and VR for museum learning, as some papers have reported common disadvantages, such as dizziness and device burden, which may be a factor influencing the learning effect. Fourth, the samples for the meta-analysis were relatively small.

There are also some issues in the existing empirical studies. First, some studies merely gave a brief background introduction without a detailed description of how AR and VR were applied in the museums (Bell & Smith, 2020). Second, the majority of studies lacked control groups as comparisons to investigate the effect of AR and VR in museum learning. Third, the true impact of museum visits may not be apparent, as the typical museum learning experience with AR and VR is only a few hours or less, and current studies mainly evaluated easily observed immediate learning outcomes of visitors/participants rather than distal outcomes such as teenagers' aspiration for science, art, etc.

6. Conclusion

This work presents a systematic literature review and meta-analysis on the contexts of using AR and VR technologies to support museum learning, design elements of how AR and VR are incorporated into the learning activities, and the effects of AR/VR-assisted museum learning on academic achievement. First, the results indicate that AR and VR have mostly been used by K-12 students in science and art museums, mainly for acquiring declarative knowledge. Second, AR and VR are often used to superimpose supplementary information onto physical exhibits, dynamically visualize typically invisible phenomena or concepts, and simulate virtual exhibition and narrative scenarios. Only a few studies have used AR and VR to improve visitors' interaction with exhibitions by allowing them to manipulate or generate virtual objects. Finally, this meta-analysis reveals that AR and VR applications have produced favorable effects on both academic achievement (ES = 0.45) and learner perceptions (ES = 0.59). Our findings may therefore contribute to a deeper understanding of AR/VR-assisted learning in museums.

Credit authorship contribution statement

Yuting Zhou: Data collection, Data analysis, and Writing– Original Draft. Juanjuan Chen: Conceptualization, Methodology, Supervision, Reviewing and Editing. Minhong Wang: Idea refinement, Evolution of overarching research goals and research questions, Reviewing and Editing.

Declaration of competing interest

None.

Acknowledgements

This project is supported by research fund from the Zhejiang Federation of Humanities and Social Sciences (No. 22NDJC005Z), National Natural Science Foundation of China (No. 61977023), Eastern Scholar Chair Professorship Fund (No. JZ2017005) from the Shanghai Municipal Education Commission of China.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.edurev.2022.100454.

References

References marked with an asterisk indicate studies included in the review.

Akçayır, M., & Akçayır, G. (2017). Advantages and challenges associated with augmented reality for education: A systematic review of the literature. *Educational Research Review, 20*, 1–11. https://doi.org/10.1016/j.edurev.2016.11.002

Alfadil, M. (2020). Effectiveness of virtual reality game in foreign language vocabulary acquisition. *Computers & Education, 153*. https://doi.org/10.1016/j.compedu.2020.103893. Article 103893.

Andre, L., Durksen, T., & Volman, M. L. (2017). Museums as avenues of learning for children: A decade of research. *Learning Environments Research, 20*(1), 47–76. https://doi.org/10.1007/s10984-016-9222-9

Arici, F., Yildirim, P., Caliklar, Ş., & Yilmaz, R. M. (2019). Research trends in the use of augmented reality in science education: Content and bibliometric mapping analysis. *Computers & Education, 142*. https://doi.org/10.1016/j.compedu.2019.103647. Article 103647.

Azuma, R. T. (1997). A survey of augmented reality. *Presence (Cambridge, Mass.), 6*(4), 355–385. https://doi.org/10.1162/pres.1997.6.4.355

Bacca, J., Baldiris, S., Fabregat, R., Graf, S., & Kinshuk. (2014). Augmented reality trends in education: A systematic review of research and applications. *Educational Technology & Society, 17*(4), 133–149.

* Baradaranrahimi, F., Boyd, J. E., Levy, R. M., & Eiserman, J. R. (2020). *New media and space: An empirical study of learning and enjoyment through museum hybrid space*. IEEE Transactions on Visualization and Computer Graphics. https://doi.org/10.1109/tvcg.2020.3043324.

Behrendt, M., & Franklin, T. (2014). A review of research on school field trips and their value in education. *International Journal of Environmental & Science Education, 9*(3), 235–245. https://doi.org/10.12973/ijese.2014.213a

* Bell, D. R., & Smith, J. K. (2020). Inside the digital learning laboratory: New directions in museum education. *Curator: The Museum Journal, 63*(3), 371–386. https://doi.org/10.1111/cura.12376.

Bettelli, A., Buson, R., Orso, V., Benvegnú, G., Pluchino, P., & Gamberini, L. (2020). Using virtual reality to enrich the visit at the botanical garden. *Annual Review of CyberTherapy and Telemedicine, 18*, 57–61.

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to meta-analysis*. Wiley. https://doi.org/10.1002/9780470743386

* Borovanska, Z., Poyade, M., Rea, P. M., & Buksh, I. D. (2020). Engaging with children using augmented reality on clothing to prevent them from smoking. In P. M. Rea (Ed.). *Biomedical visualisation* (Vol. 7, pp. 59–94). https://doi.org/10.1007/978-3-030-43961-3_4. Springer International Publishing.

* Bossavit, B., Pina, A., Sanchez-Gil, I., & Urtasun, A. (2018). Educational games to enhance museum visits for schools. *Educational Technology & Society, 21*(4), 171–186.

Cai, S., Chiang, F. K., & Wang, X. (2013). Using the augmented reality 3D technique for a convex imaging experiment in a physics course. *International Journal of Engineering Education, 29*(4), 856–865.

Çakiroğlu, Ü., & Gökoğlu, S. (2019). Development of fire safety behavioral skills via virtual reality. *Computers & Education, 133*, 56–68. https://doi.org/10.1016/j.compedu.2019.01.014

Calle-Bustos, A. M., Juan, M. C., García-García, I., & Abad, F. (2017). An augmented reality game to support therapeutic education for children with diabetes. *PLoS One, 12*(9), Article e0184645. https://doi.org/10.1371/journal.pone.0184645

Carmigniani, J., Furht, B., Anisetti, M., Ceravolo, P., Damiani, E., & Ivkovic, M. (2011). Augmented reality technologies, systems and applications. *Multimedia Tools and Applications, 51*(1), 341–377. https://doi.org/10.1007/s11042-010-0660-6

* Chang, K.-E., Chang, C.-T., Hou, H.-T., Sung, Y.-T., Chao, H.-L., & Lee, C.-M. (2014). Development and behavioral pattern analysis of a mobile guide system with augmented reality for painting appreciation instruction in an art museum. *Computers & Education, 71*, 185–197. https://doi.org/10.1016/j.compedu.2013.09.022.

Chen, J., Wang, M., Kirschner, P. A., & Tsai, C. C. (2018). The role of collaboration, computer use, learning environments, and supporting strategies in CSCL: A meta-analysis. *Review of Educational Research, 88*(6), 799–843. https://doi.org/10.3102/0034654318791584

Chiang, C., Wells, P. K., & Xu, G. (2021). How does experiential learning encourage active learning in auditing education? *Journal of Accounting Education, 54*, Article 100713. https://doi.org/10.1016/j.jaccedu.2020.100713

Chiu, J. L., DeJaegher, C. J., & Chao, J. (2015). The effects of augmented virtual science laboratories on middle school students' understanding of gas properties. *Computers & Education, 85*, 59–73. https://doi.org/10.1016/j.compedu.2015.02.007

Choi, H.-G., & Ahn, S. H. (2021). Effects of a conflict resolution training program on nursing students: A quasi-experimental study based on the situated learning theory. *Nurse Education Today, 103*. https://doi.org/10.1016/j.nedt.2021.104951. Article 104951.

* Connaghan, R., Poyade, M., & Rea, P. M. (2019). Evaluation of child-friendly augmented reality tool for patient-centered education in radiology and bone reconstruction. In P. M. Rea (Ed.). *Biomedical visualisation* (Vol. 4, pp. 105–126). https://doi.org/10.1007/978-3-030-24281-7_9. Springer International Publishing.

Crowley, K., Pierroux, P., & Knutson, K. (2014). Informal learning in museums. In R. K. Sawyer (Ed.), *The cambridge handbook of the learning sciences* (2nd ed. ed., pp. 461–478). Cambridge University Press. https://doi.org/10.1017/CBO9781139519526.028.

* Damala, A., Hornecker, E., van der Vaart, M., van Dijk, D., & Ruthven, I. (2016). The loupe: Tangible augmented reality for learning to look at ancient Greek art. *Mediterranean Archaeology and Archaeometry, 16*(5), 73–85. https://doi.org/10.5281/zenodo.204970.

Danaei, D., Jamali, H. R., Mansourian, Y., & Rastegarpour, H. (2020). Comparing reading comprehension between children reading augmented reality and print storybooks. *Computers & Education, 153*. https://doi.org/10.1016/j.compedu.2020.103900. Article 103900.

De Rijcke, S., & Beaulieu, A. (2011). Image as interface: Consequences for users of museum knowledge. *Library Trends, 59*(4), 663–685. https://doi.org/10.1353/lib.2011.0020

Dieck, M. C. T., Jung, T. H., & Dieck, D. T. (2018). Enhancing art gallery visitors' learning experience using wearable augmented reality: Generic learning outcomes perspective. *Current Issues in Tourism, 21*(17), 2014–2034. https://doi.org/10.1080/13683500.2016.1224818

* Dudzik, B. (2018). Visitor perceptions of augmented reality in science museums. *Master's Thesis*. Ann Arbor: University of Washington. Publication Number 10828096).

Falk, J. H., & Dierking, L. D. (2000). *Learning from museums: Visitor experiences and the making of meaning.* AltaMira Press.

Falomo Bernarduzzi, L., Bernardi, E. M., Ferrari, A., Garbarino, M. C., & Vai, A. (2021). Augmented reality application for handheld devices: How to make it hAPPen at the pavia university history museum. *Science & Education, 30*(3), 755–773. https://doi.org/10.1007/s11191-021-00197-z

Garzón, J., Pavón, J., & Baldiris, S. (2019). Systematic review and meta-analysis of augmented reality in educational settings. *Virtual Reality : The Journal of the Virtual Reality Society, 23*(4), 447–459. https://doi.org/10.1007/s10055-019-00379-9

* Ghouaiel, N., Cieutat, J.-M., Jessel, J.-P., & Garbaya, S. (2017). Mobile augmented reality in museums: Towards enhancing visitor's learning experience. *International Journal of Virtual Reality, 17*(1), 21–31. https://doi.org/10.20870/IJVR.2017.17.1.2885.

Goff, E. E., Mulvey, K. L., Irvin, M. J., & Hartstone-Rose, A. (2018). Applications of augmented reality in informal science learning sites: A review. *Journal of Science Education and Technology, 27*(5), 433–447. https://doi.org/10.1007/s10956-018-9734-4

* Guazzaroni, G. (2013). Emotional mapping of the archaeologist game. *Computers in Human Behavior, 29*(2), 335–344. https://doi.org/10.1016/j.chb.2012.06.008.

Guzin, O. A., Yildirim, R. G., & Ellez, A. M. (2017). An alternative educational method in early childhood: Museum education. *Educational Research and Reviews, 12*(14), 688–694. https://doi.org/10.5897/err2017.3145

* Hammady, R., Ma, M., & Strathearn, C. (2020). Ambient information visualisation and visitors technology acceptance of mixed reality in museums. *Journal on Computing and Cultural Heritage, 13*(2), 1–22. https://doi.org/10.1145/3359590.

Harrington, M. C. R., Tatzgern, M., Langer, T., & Wenzel, J. W. (2019). Augmented reality brings the real world into natural history dioramas with data visualizations and bioacoustics at the Carnegie Museum of Natural History. *Curator: The Museum Journal, 62*(2), 177–193. https://doi.org/10.1111/cura.12308

* Haryani, P., & Triyono, J. (2020). The designing of interactive learning media at Yogyakarta's sandi museum based on augmented reality. *International Journal on Informatics Visualization, 4*(1), 52–57. https://doi.org/10.30630/joiv.4.1.157.

Hassan, A., & Ramkissoon, H. (2016). Augmented reality application to museum visitor experiences. In J. N. Albrecht (Ed.), *Visitor management in tourism destinations* (pp. 117–130). CABI.

* Hsiao, H. S., Chang, C. S., Lin, C. Y., & Wang, Y. Z. (2016). Weather observers: A manipulative augmented reality system for weather simulations at home, in the classroom, and at a museum. *Interactive Learning Environments, 24*(1), 205–223. https://doi.org/10.1080/10494820.2013.834829.

* Hsu, T. Y., Liang, H., Chiou, C. K., & Tseng, J. C. R. (2018). CoboChild: A blended mobile game-based learning service for children in museum contexts. *Data Technologies and Applications, 52*(3), 294–312. https://doi.org/10.1108/DTA-05-2016-0042.

Huang, T.-C., Chen, C.-C., & Chou, Y.-W. (2016). Animating eco-education: To see, feel, and discover in an augmented reality-based experiential learning environment. *Computers & Education, 96*, 72–82. https://doi.org/10.1016/j.compedu.2016.02.008

Ibanez, M.-B., & Delgado-Kloos, C. (2018). Augmented reality for STEM learning: A systematic review. *Computers & Education, 123*, 109–123. https://doi.org/10.1016/j.compedu.2018.05.002

ICOM. (2007). *Museum definition-ICOM*. https://icom.museum/en/resources/standards-guidelines/museum-definition/.

Innocenti, E. D., Geronazzo, M., Vescovi, D., Nordahl, R., Serafin, S., Ludovico, L. A., & Avanzini, F. (2019). Mobile virtual reality for musical genre learning in primary education. *Computers & Education, 139*, 102–117. https://doi.org/10.1016/j.compedu.2019.04.010

Johnson, E. B. (2002). *Contextual teaching and learning: What it is and why it's here to stay.* Corwin Press.

Juan, M., Loachamín-Valencia, M., Garcia-Garcia, I., Melchor, J. M., & Benedito, J. (2017). ARCoins. An augmented reality app for learning about numismatics. *2017 IEEE 17th International Conference on Advanced Learning Technologies (ICALT)*. https://doi.org/10.1109/ICALT.2017.27.

* Jung, T., tom Dieck, M. C., Lee, H., & Chung, N. (2016). Effects of virtual reality and augmented reality on visitor experiences in museum. *Information and Communication Technologies in Tourism*, 621–635. https://doi.org/10.1007/978-3-319-28231-2_45, 2016.

* Khan, M. A., Israr, S., Almogren, A. S., Din, I. U., Almogren, A., & Rodrigues, J. J. P. C. (2020). Using augmented reality and deep learning to enhance Taxila Museum experience. *Journal of Real-Time Image Processing, 18*(2), 321–332. https://doi.org/10.1007/s11554-020-01038-y.

Kisiel, J. F. (2003). Teachers, museums and worksheets: A closer look at a learning experience. *Journal of Science Teacher Education, 14*(1), 3–21. https://doi.org/10.1023/A:1022991222494

Kisiel, J. (2006). Making field trips work. *The Science Teacher, 73*(1), 46–48.

Kolb, D. A. (2014). *Experiential learning: Experience as the source of learning and development* (2nd ed.). Pearson Education, Inc.

* Kosa, T., Bennett, L., Livingstone, D., Goodyear, C., & Loranger, B. (2019). Innovative education and engagement tools for rheumatology and immunology public engagement with augmented reality. In P. M. Pea (Ed.), *5. Biomedical visualisation* (pp. 105–116). Springer International Publishing https://doi.org/10.1007/978-3-030-31904-5_8.

* Koutsabasis, P., & Vosinakis, S. (2018). Kinesthetic interactions in museums: Conveying cultural heritage by making use of ancient tools and (re-) constructing artworks. *Virtual Reality : The Journal of the Virtual Reality Society, 22*(2), 103–118. https://doi.org/10.1007/s10055-017-0325-0.

Kyndt, E., Raes, E., Lismont, B., Timmers, F., Cascallar, E., & Dochy, F. (2013). A meta-analysis of the effects of face-to-face cooperative learning. Do recent studies falsify or verify earlier findings? *Educational Research Review, 10*, 133–149. https://doi.org/10.1016/j.edurev.2013.02.002

Lewalter, D., Gegenfurtner, A., & Renninger, K. A. (2021). Out-of-school programs and interest: Design considerations based on a meta-analysis. *Educational Research Review, 34*. https://doi.org/10.1016/j.edurev.2021.100406. Article 100406.

* Li, P., & Chang, P. (2017). A study of virtual reality experience value and learning efficiency of museum - using Shihsanhang museum as an example. In None (Ed.), *2017 International Conference on Applied System Innovation (ICASI)* (pp. 1158–1161) https://doi.org/10.1109/ICASI.2017.7988391.

Lin, C. J., Tang, K. Y., & Tu, Y. F. (2021). *Advancements and research topics of museum-based mobile learning: A review of journal publications from 2008 to 2019* (pp. 1–24). Interactive Learning Environments. In press https://doi.org/10.1080/10494820.2021.1981394.

Lipsey, M. W., & Wilson, D. B. (2001). *Practical meta-analysis.* Sage Publications, Inc.

Makransky, G., & Lilleholt, L. (2018). A structural equation modeling investigation of the emotional value of immersive virtual reality in education. *Educational Technology Research & Development, 66*(5), 1141–1164. https://doi.org/10.1007/s11423-018-9581-2

* Marín-Morales, J., Higuera-Trujillo, J. L., De-Juan-Ripoll, C., Llinares, C., Guixeres, J., Iñarra, S., & Alcañiz, M. (2019). Navigation comparison between a real and a virtual museum: Time-dependent differences using a head mounted display. *Interacting with Computers, 31*(2), 208–220. https://doi.org/10.1093/iwc/iwz018.

Mayer, R. E. (2014). Cognitive theory of multimedia learning. In *The cambridge handbook of multimedia learning* (2nd ed., pp. 43–71). https://doi.org/10.1017/CBO9781139547369.005

Mayer, R. E. (2020). Searching for the role of emotions in e-learning. *Learning and Instruction, 70*. https://doi.org/10.1016/j.learninstruc.2019.05.010. Article 101213.

Mayer, R. E., & Estrella, G. (2014). Benefits of emotional design in multimedia instruction. *Learning and Instruction, 33*, 12–18. https://doi.org/10.1016/j.learninstruc.2014.02.004

Moher, D., Liberati, A., Tetzlaff, J., & Altman, D. G. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *Journal of Clinical Epidemiology, 62*(10), 1006–1012. https://doi.org/10.1016/j.jclinepi.2009.06.005

* Moorhouse, N., tom Dieck, M. C., & Jung, T. (2019). An experiential view to children learning in museums with Augmented Reality. *Museum Management and Curatorship, 34*(4), 402–418. https://doi.org/10.1080/09647775.2019.1578991.

Morélot, S., Garrigou, A., Dedieu, J., & N'Kaoua, B. (2021). Virtual reality for fire safety training: Influence of immersion and sense of presence on conceptual and procedural acquisition. *Computers & Education, 166*. https://doi.org/10.1016/j.compedu.2021.104145. Article 104145.

* Nechita, F., & Rezeanu, C. I. (2019). Augmenting museum communication services to create young audiences. *Sustainability, 11*(20), 5830. https://doi.org/10.3390/su11205830.

* Oh, S., So, H.-J., & Gaydos, M. (2018). Hybrid augmented reality for participatory learning: The hidden efficacy of multi-user game-based simulation. *IEEE Transactions on Learning Technologies, 11*(1), 115–127. https://doi.org/10.1109/TLT.2017.2750673.

Pellas, N., Fotaris, P., Kazanidis, I., & Wells, D. (2019). Augmenting the learning experience in primary and secondary school education: A systematic review of recent trends in augmented reality game-based learning. *Virtual Reality: The Journal of the Virtual Reality Society, 23*(4), 329–346. https://doi.org/10.1007/s10055-018-0347-2

* Poce, A., Amenduni, F., de Medio, C., Valente, M., & Re, M. R. (2019). Adopting augmented reality to engage higher education students in a museum university collection: The experience at Roma Tre University. *Information, 10*(12), 373. https://doi.org/10.3390/info10120373.

Prince, M. (2004). Does active learning work? A review of the research. *Journal of Engineering Education, 93*(3), 223–231. https://doi.org/10.1002/j.2168-9830.2004.tb00809.x

* Puig, A., Rodríguez, I., Arcos, J. L., Rodríguez-Aguilar, J. A., Cebrián, S., Bogdanovych, A., & Piqué, R. (2020). Lessons learned from supplementing archaeological museum exhibitions with virtual reality. *Virtual Reality: The Journal of the Virtual Reality Society, 24*(2), 343–358. https://doi.org/10.1007/s10055-019-00391-z.

Radianti, J., Majchrzak, T. A., Fromm, J., & Wohlgenannt, I. (2020). A systematic review of immersive virtual reality applications for higher education: Design elements, lessons learned, and research agenda. *Computers & Education, 147*. https://doi.org/10.1016/j.compedu.2019.103778. Article 103778.

Renganayagalu, S., Mallam, S. C., & Nazir, S. (2021). *Effectiveness of VR head mounted displays in professional training: A systematic review*. https://doi.org/10.1007/s10758-020-09489-9. Technology, Knowledge and Learning.

* Rhee, B. (2019). An analysis of information and communication technology and virtual reality technology implementation through a quantitative research on users' experiences. *Journal of Theoretical and Applied Information Technology, 97*(18), 4797–4810.

* Salmi, H., Thuneberg, H., & Vainikainen, M. P. (2017). Making the invisible observable by augmented reality in informal science education context. *International Journal of Science Education, Part B: Communication and Public Engagement, 7*(3), 253–268. https://doi.org/10.1080/21548455.2016.1254358.

Saltan, F., & Arslan, O. (2017). The use of augmented reality in formal education: A scoping review. *Eurasia Journal of Mathematics, Science and Technology Education, 13*(2), 503–520. https://doi.org/10.12973/eurasia.2017.00628a

* Savela, N., Oksanen, A., Kaakinen, M., Noreikis, M., & Xiao, Y. (2020). Does augmented reality affect sociability, entertainment, and learning? A field experiment. *Applied Sciences, 10*(4). https://doi.org/10.3390/app10041392. Article 1392.

Scavarelli, A., Arya, A., & Teather, R. J. (2021). Virtual reality and augmented reality in social learning spaces: A literature review. *Virtual Reality: The Journal of the Virtual Reality Society, 25*(1), 257–277. https://doi.org/10.1007/s10055-020-00444-8

Schwan, S., Grajal, A., & Lewalter, D. (2014). Understanding and engagement in places of science experience: Science museums, science centers, zoos, and aquariums. *Educational Psychologist, 49*(2), 70–85. https://doi.org/10.1080/00461520.2014.917588

Smith, J. W. (2015). Immersive virtual environment technology to supplement environmental perception, preference and behavior research: A review with applications. *International Journal of Environmental Research and Public Health, 12*(9), 11486–11505. https://doi.org/10.3390/ijerph120911486

* Sommerauer, P., & Müller, O. (2014). Augmented reality in informal learning environments: A field experiment in a mathematics exhibition. *Computers & Education, 79*, 59–68. https://doi.org/10.1016/j.compedu.2014.07.013.

* Sugiura, A., Kitama, T., Toyoura, M., & Mao, X. (2019). The use of augmented reality technology in medical specimen museum tours. *Anatomical Sciences Education, 12*(5), 561–571. https://doi.org/10.1002/ase.1822.

* Sulaiman, S., Ab Aziz, N. H., Adzmi, S. A. M., Ihsan, N. A. S., Jaidi, N. I., Yaziz, A. F. A., & Misroom, M. I. (2019). *Museum informatics: A case study on augmented reality at tanjung balau fishermen museum*, 2019 IEEE 9th International Conference on System Engineering and Technology (ICSET). https://doi.org/10.1109/ICSEngT.2019.8906341.

Sultan, L., Abuznadah, W., Al-Jifree, H., Anwar Khan, M., Alsaywid, B., & Ashour, F. (2019). An experimental study on usefulness of virtual reality 360° in undergraduate medical education. *Advances in Medical Education and Practice, 10*, 907–916. https://doi.org/10.2147/AMEP.S219344

* Tabone, W. (2020). The effectiveness of an augmented reality guiding system in an art museum. In D. Seychell, & A. Dingli (Eds.), *Rediscovering heritage through technology: A collection of innovative research case studies that are reworking the way we experience heritage* (pp. 197–214). Springer International Publishing. https://doi.org/10.1007/978-3-030-36107-5_10.

* Takahashi, T. B., Takahashi, S., Kusunoki, F., Terano, T., & Inagaki, S. (2013). *Making a hands-on display with augmented reality work at a science museum. 2013 International Conference on Signal-Image Technology & Internet-Based Systems*. https://doi.org/10.1109/SITIS.2013.69.

Tanner-Smith, E. E., & Tipton, E. (2014). Robust variance estimation with dependent effect sizes: Practical considerations including a software tutorial in stata and SPSS. *Research Synthesis Methods, 5*(1), 13–30. https://doi.org/10.1002/jrsm.1091

* Wang, J. S. (2014). *The impact of multiple dynamic visualizations on family children's learning in a science museum*. Ann Arbor: D., University of Pennsylvania (Publication Number 3622675).

Wang, H. Y., Liu, G. Z., & Hwang, G. J. (2017). Integrating socio-cultural contexts and location-based systems for ubiquitous language learning in museums: A state of the art review of 2009-2014. *British Journal of Educational Technology, 48*(2), 653–671. https://doi.org/10.1111/bjet.12424

Watson, B., & Werb, S. R. (2013). One hundred strong: A colloquium on transforming natural history museums in the twenty-first century. *Curator: The Museum Journal, 56*(2), 255–265. https://doi.org/10.1111/cura.12023

Xanthoudaki, M. (1998). Is it always worth the trip? The contribution of museum and gallery educational programmes to classroom art education. *Cambridge Journal of Education, 28*(2), 181–195. https://doi.org/10.1080/0305764980280204

Xu, W., Dai, T. T., Shen, Z. Y., & Yao, Y. J. (1998). Effects of technology application on museum learning: A meta-analysis of 42 studies published between 2011 and 2021. *Interactive Learning Environments*, 1–16. https://doi.org/10.1080/10494820.2021.1976803.

* Yoon, S. A., Anderson, E., Lin, J., & Elinich, K. (2017). How augmented reality enables conceptual understanding of challenging science content. *Educational Technology & Society, 20*(1), 156–168.

* Yoon, S. A., Anderson, E., Park, M., Elinich, K., & Lin, J. (2018). How augmented reality, textual, and collaborative scaffolds work synergistically to improve learning in a science museum. *Research in Science & Technological Education, 36*(3), 261–281. https://doi.org/10.1080/02635143.2017.1386645.

* Yoon, S. A., Elinich, K., Wang, J., Steinmeier, C., & Tucker, S. (2012). Using augmented reality and knowledge-building scaffolds to improve learning in a science museum. *International Journal of Computer-Supported Collaborative Learning, 7*(4), 519–541. https://doi.org/10.1007/s11412-012-9156-x.

* Yoon, S. A., Elinich, K., Wang, J., Van Schooneveld, J. B., & Anderson, E. (2013). Scaffolding informal learning in science museums: How much is too much?. *Science & Education, 97*(6), 848–877. https://doi.org/10.1002/sce.21079.

* Yoon, S. A., & Wang, J. (2014). Making the invisible visible in science museums through augmented reality devices. *TechTrends, 58*(1), 49–55. https://doi.org/10.1007/s11528-013-0720-7.

* Yu, S. J., Sun, J. C. Y., & Chen, O. T. C. (2019). Effect of AR-based online wearable guides on university students' situational interest and learning performance. *Universal Access in the Information Society, 18*(2), 287–299. https://doi.org/10.1007/s10209-017-0591-3.

Yuting Zhou is a graduate student in the College of Education, Zhejiang University.

Juanjuan Chen is a research professor in the College of Education, Zhejiang University.

Minhong Wang is a full professor in the Faculty of Education, The University of Hong Kong.